

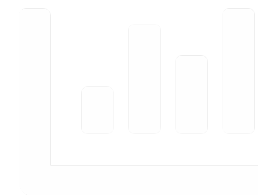
Topic 4



Lecture 4 · Probability

National University of Singapore (NUS)

TODAY'S OBJECTIVES



- 1 Introduction
- 2 Sample Space and Events
- 3 Axioms of Probability and Basic Properties
- 4 Independence
- 5 Conditional Probability
- 6 Law of Total Probability and Bayes' Theorem

Why do we need probability?

- Making decisions when uncertain about the outcome
- Possible outcomes are known, but uncertain which outcome will occur for any given observation
 - Should we invest in the stock market?
 - Should we start a new business?
 - Should we carry an umbrella in case it rains?
- Probability enables us to quantify uncertainty and risk in any decision
 - Helps use to make a good decision (hopefully)

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Sample Space and Events

Sample Space

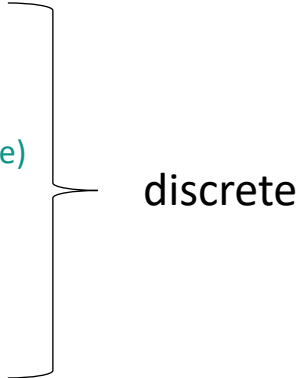


DEFINITION

Sample space is the set of all possible outcomes of a random phenomenon. It is usually denoted by the letter S .

- The same experiment may have different sample spaces
- Not all outcomes in a sample space are equally likely to happen

Examples of Sample Space

- Toss a coin twice:
 - Outcome: the side on top for each toss
 - $S = \{(\text{Heads Heads}), (\text{Heads Tails}), (\text{Tails Heads}), (\text{Tails Tails})\}$ (basic sample space)
 - Toss a coin twice:
 - Outcome: the number of Heads
 - $S = \{0, 1, 2\}$
- 
- discrete
- Basic sample space helps us to have other sample spaces for the same experiment
 - Relative humidity of air on a random day at a random location:
 - $S = [0, 100]$

Event



DEFINITION

An event is a subset of the sample space, S .

An event corresponds to a particular outcome, or group of outcomes.

EXAMPLE

- Roll a six-sided fair dice:
 - Outcome: the number on top
 - $S = \{1, 2, 3, 4, 5, 6\}$
 - Event: outcome is an even number, ie. $\{2, 4, 6\}$

EXAMPLE

- Toss a fair coin once:
 - Outcome: the side on top
 - $S = \{\text{Heads}, \text{Tails}\}$
 - Event: outcome is Heads, ie. $\{\text{Heads}\}$

Examples of Events

- Toss a coin twice:
 - Outcome: the side on top for each toss
 - $S = \{(\text{Heads Heads}), (\text{Heads Tails}), (\text{Tails Heads}), (\text{Tails Tails})\}$ (basic sample space)
 - Event: outcome is exactly one Heads, ie. $\{(\text{Heads Tails}), (\text{Tails Heads})\}$
 - Toss a coin twice:
 - Outcome: the number of Heads
 - $S = \{0, 1, 2\}$
 - Event: outcome is exactly one Heads, ie. $\{1\}$
- 
- discrete
- Basic sample space helps us to have other sample spaces for the same experiment
 - Relative humidity of air on a random day at a random location:
 - $S = [0, 100]$
 - Event: measurement is at least 90, ie. $[90, 100]$

Union



DEFINITION

The union of events A and B is the event that consists of all outcomes that belong to A or B or both.

NOTATION

$A \cup B$: union of A and B

EXAMPLE

- Roll a six-sided fair dice:
 - Event A: outcome is even, ie. {2, 4, 6}
 - Event B: outcome is at least 3, ie. {3, 4, 5, 6}
 - Event $A \cup B$: outcome is even or at least 3, ie. {2, 3, 4, 5, 6}

Intersection



DEFINITION

The intersection of events A and B is the event that consists of all outcomes common to both A and B.

NOTATION

$A \cap B$: intersection of A and B

EXAMPLE

- Roll a six-sided fair dice:
 - Event A: outcome is even, ie. {2, 4, 6}
 - Event B: outcome is at least 3, ie. {3, 4, 5, 6}
 - Event $A \cap B$: outcome is an even number that is at least 3, ie. {4, 6}

Complement



DEFINITION

The complement of an event A is the event that consists of all outcomes in the sample space that are not in A .

NOTATION

A^c : complement of A

A' : complement of A

EXAMPLE

- Roll a six-sided fair dice:
 - Event A : outcome is even, ie. $\{2, 4, 6\}$
 - Event A^c : outcome is odd, ie. $\{1, 3, 5\}$

Mutually Exclusive



DEFINITION

Mutually exclusive events A and B, or disjoint events, do not have any common outcome. Their intersection is the empty set. These events cannot occur simultaneously.

NOTATION

$$A \cap B = \emptyset$$

EXAMPLE

- $A \cap A^c = \emptyset$
- Roll a six-sided fair dice:
 - Event C: outcome is at most 3, ie. {1, 2, 3}
 - Event D: outcome is larger than 4, ie. {5, 6}
 - $C \cap D = \emptyset$

Examples of Event Algebra

- Roll two six-sided fair die:
 - Outcome: sum of the values on the die
 - $S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
 - Event A: outcome is a multiple of 3, ie. $\{3, 6, 9, 12\}$
 - Event B: outcome is a multiple of 4, ie. $\{4, 8, 12\}$
 - Event $A \cup B$: outcome is either a multiple of 3 or 4, ie. $\{3, 4, 6, 8, 9, 12\}$
 - Event A^c : outcome is not a multiple of 3, ie. $\{2, 4, 5, 7, 8, 10, 11\}$
 - Event C: outcome is even, ie. $\{2, 4, 6, 8, 10, 12\}$
 - Event D: outcome is odd, ie. $\{3, 5, 7, 9, 11\}$
 - Event C and D are mutually exclusive, ie. $C \cap D = \emptyset$
 - Event C and D are complement of each other
 - Event $A \cap B \cap C$: outcome is even and a multiple of 3 and 4, ie. $\{12\}$

Probability



DEFINITION

The probability of an event is the proportion of times that this event occurs, in a long run of trials.

- A probability is always between 0 and 1
- We need a large number of experiments to determine probability

Long Run of Trials

- Toss a fair coin once and obtain Heads
 - Would you conclude that probability of getting Heads is 100%? **No!**
- Sample 10 students and all are right-handed
 - Would you conclude that probability of being right-handed is 100%? **No!**
- Roll a fair six-sided dice and obtain 1
 - Probability = $\frac{1}{6}$

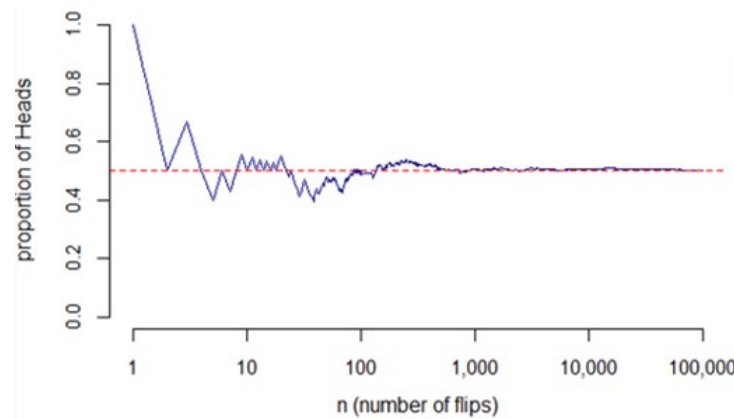
Long Run of Trials

- Toss a fair coin
- Probability of obtaining Heads = 0.5
- $S = \{\text{Heads, Tails}\}$
- Event $A = \{\text{Heads}\}$ $P(A)$
- Toss the coin n times, under the same conditions
- Consider the proportion of Heads in n tosses, for $n = 1, \dots, 10\,000$

Long Run of Trials

- When n is small, proportion of Heads is unstable
- As the number of experiments increases, the value of proportion settles down
- The number that it settles down to is the probability of the event in the long run

n	Outcomes	Proportion of Heads
1	Heads	$\frac{1}{1} = 1$
2	Heads, Tails	$\frac{1}{2} = 0.5$
3	Heads, Tails, Heads	$\frac{2}{3} = 0.667$



03

Axioms of Probability and Basic Properties

Axioms of Probability

- S : sample space, A is an event within S with probability $P(A)$

AXIOM 1

$$P(A) \geq 0$$

AXIOM 2

$P(S) = 1$ since S includes all possible outcomes of an experiment

AXIOM 3

$P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive

Axioms of Probability

- S : sample space, A is an event within S with probability $P(A)$

AXIOM 4

$P(A_1 + A_2 + \dots + A_R) = P(A_1) + P(A_2) + \dots + P(A_R)$ if $A_1, A_2, \dots, A_R$ are pairwise mutually exclusive (no two of these events can occur simultaneously)

AXIOM 5

Additive Law of Probability: $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

Equally Likely Outcomes

- When all outcomes are equally likely to happen, then $P(A) = \text{number of outcomes in } A \div \text{total number of outcomes in } S$
- Toss a fair coin:
 - $S = \{\text{Heads, Tails}\}$
 - Event A: outcome is Tails
 - $P(A) = 0.5$
- Select a card from a well-shuffled deck of 52 cards
 - S has 52 outcomes
 - Event B: outcome is a card with a heart (13 outcomes)
 - $P(B) = \frac{1}{4}$

Equally Likely Outcomes

- Assume giving birth to a boy or girl is equally likely
- A couple decide to have 3 children
 - $S = \{BBB, BBG, BGB, GBB, BGG, GBG, GGB, GGG\}$
 - Event A: at least one child of each gender, ie. $\{BBG, BGB, GBB, BGG, GBG, GGB\}$
 - $P(A) = 0.75$
 - Event B: two boys and one girl, ie. $\{BBG, BGB, GBB\}$
 - $P(B) = \frac{3}{8}$

Equally Likely Outcomes

- 6 people are queueing for a bus,
 - Two of them named Jack and Jill
 - Assume all orderings are equally likely
- Event A: Jack is right in front of Jill, with no one between them
- $S = 6! = 720$
- $A = 5! = 120$
- $P(A) = \frac{120}{720} = \frac{1}{6}$

Implications of the Axioms

- $P(A^c) = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A) = P(A \cap B) + P(A \cap B^c)$

Example 1: Cocaine Users

- Data collected in the US on cocaine usage
- $P(M) = P(M \cap A) + P(M \cap B) + P(M \cap C) = 0.29 + 0.16 + 0.23 = 0.68$
- $P(F) = 1 - P(M) = 1 - 0.68 = 0.32$

Lifetime Use	Male	Female
1 – 19 times (A)	0.29	0.06
20 – 99 times (B)	0.16	0.18
100+ times (C)	0.23	0.08

Example 1: Cocaine Users

- $P(M \text{ or } C) = P(M \cup C) = P(M) + P(C) - P(M \cap C) = 0.68 + (0.23 + 0.08) - 0.23 = 0.76$
- Events are of unequally likely outcomes
 - When randomly sampling an individual, the outcome on gender are not equally likely
 - When randomly sampling an individual, the outcome on the level of usage are not equally likely

Unequally Likely Outcomes

METHOD 1

- Easiest way
- Transform the sample space to an equivalent sample space that has equally likely outcomes
- Probability of events can be calculated easily

METHOD 2

- Repeat the experiment many times until the proportion needed settles down
- Usually not possible in reality
- Instead, repeat the experiment several times and estimate the true proportion

METHOD 3

- May have reason to believe outcomes have particular probabilities from experience
- Directly assign these probabilities to the individual outcomes
- Should have prior experience

Unequally Likely Outcomes – Method 1

- Toss a fair coin twice
- Outcome: number of Heads appearing
- $S = \{0, 1, 2\}$, ie. $\{(\text{Heads Tails}), (\text{Heads Heads}), (\text{Tails Heads}), (\text{Tails Tails})\}$
- Event A: obtain 1 Head in of 2 tosses, ie. $\{(\text{Heads Tails}), (\text{Tails Heads})\}$
- $P(A) = \frac{2}{4} = 0.5$

Example 2: Diastolic Blood Pressure (DBP)

- $S = [50, 140]$
 - Event A: DBP is normal, ie. $70 \leq \text{DBP} \leq 90$
 - Event B: DBP is borderline high, ie. $90 < \text{DBP} \leq 95$
- } disjoint
- Suppose we know 70% of people have normal DBP, 10% of people have borderline high DBP
 - Event Z: DBP is from 70 to 95
 - $P(Z) = P(A \cup B) = 0.8$

04

Independence

Independence



DEFINITION

Events A and B are independent if $P(A \cap B) = P(A)P(B)$

Two events that are not independent are said to be dependent.

- Independence between two events implies that these two events do not influence each other
- Knowing that A happened (or did not happen) does not say anything about whether B happened (or did not happen)
- Events from different experiments are independent

Independence

DIRECTION 1

- To check if events are independent
- Given $P(A)$, $P(B)$
- Compute $P(A \cap B)$
- Does $P(A \cap B) = P(A)P(B)$?

DIRECTION 2

- Assume events are independent
- Given $P(A)$, $P(B)$
- Compute $P(A \cap B)$ using $P(A \cap B) = P(A)P(B)$

Independence – Direction 2

PROBLEM

We toss a coin twice.

Event A_1 : we obtain Heads
Event A_2 : we obtain Tails

} independent

GIVEN

- $P(A_1) = 0.5$
- $P(A_2) = 0.5$

FIND

- $P(A_1 \cap A_2) = P(A_1)P(A_2) = 0.5 \times 0.5 = 0.25$

Independence – Direction 1

PROBLEM

Draw a card from a well-shuffled deck of 52 cards.

Event A: card drawn is a Clubs

Event B: card drawn is a picture Card } independent

Event $A \cap B$: card drawn is a Clubs and a picture card

GIVEN

- $P(A) = \frac{13}{52} = \frac{1}{4}$
- $P(B) = \frac{12}{52} = \frac{3}{13}$

FIND

- $P(A \cap B) = \frac{3}{32}$  Equal
- $P(A)P(B) = \frac{1}{4} \times \frac{3}{13} = \frac{3}{52}$

Dependent Events

PROBLEM

We roll a six-sided fair dice. $S = \{1, 2, 3, 4, 5, 6\}$

Event A: outcome is even

Event B: outcome is either 3, 4 or 6

} Dependent!

GIVEN

- $P(A) = 0.5$
- $P(B) = 0.5$

FIND

- $A \cap B = \{4, 6\}$
- $P(A \cap B) = \frac{2}{6} = \frac{1}{3} \neq P(A)P(B)$

Mutually Exclusive Events and Independence

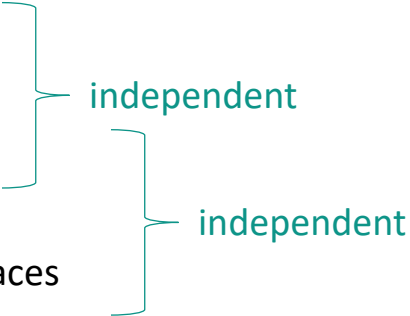
- Events A and B are mutually exclusive
 - If A occurs, we know B cannot occur (since they are disjoint)
 - Are they independent? **No!**
 - If $P(A) > 0$ and $P(B) > 0$, then $P(A \cap B) = 0$
 - But $P(A)P(B) \neq 0$



KEY INSIGHT

Mutually exclusive implies dependence!

Mutually Exclusive Events and Independence

- Event W: Joe Wrigley is walking
 - Event C: Joe Wrigley is chewing gum
 - Event T: Joe Wrigley is tying his shoelaces
 - W and T are mutually exclusive
- 
- The diagram consists of three teal-colored curly braces. The first brace groups 'Event W: Joe Wrigley is walking' and 'Event C: Joe Wrigley is chewing gum', with the word 'independent' in teal text to its right. The second brace groups 'Event C: Joe Wrigley is chewing gum' and 'Event T: Joe Wrigley is tying his shoelaces', also with the word 'independent' in teal text to its right. This indicates that W and C are independent, and C and T are independent. The fourth bullet point states that W and T are mutually exclusive.

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Conditional Probability

Conditional Probability

FORMULA

Events A and B are within a sample space S .

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

when $P(B) > 0$



KEY INSIGHT

The numerator is always the intersection of both events.



KEY INSIGHT

The denominator is always the given event.

Conditional Probability

- Draw a card from well-shuffled deck of 52 cards
- Event A: card is Clubs
 - $P(A) = \frac{13}{52} = \frac{1}{4}$
- A card is drawn and we are told it is black
- Event B: card is black
 - $P(B) = 0.5$
- Event $A \cap B$: card is Clubs and black
 - A is a subset of B
 - $P(A \cap B) = P(A) = \frac{1}{4}$
- $$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2} > \frac{1}{4} = P(A)$$

Example 3: Triple Blood Test

- Screening test for pregnant women to estimate the risk of a baby being born with Down Syndrome
- Assume this sample is presentative of the population
- Event POS: test is positive
 - $P(\text{POS}) = \frac{1355}{5282}$

DS Status	Test Result		
	NEG	POS	Total
No DS	3921	1307	5228
DS	6	48	54
Total	3927	1355	5282

Example 3: Triple Blood Test

- Screening test for pregnant women to estimate the risk of a baby being born with Down Syndrome
- Assume this sample is presentative of the population
- Event POS: test is positive, Event DS: Down Syndrome present
 - $P(\text{DS}|\text{POS}) = \frac{48}{1355}$

DS Status	Test Result		
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No DS	3921	1307	5228
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Example 3: Triple Blood Test

- Screening test for pregnant women to estimate the risk of a baby being born with Down Syndrome
- Assume this sample is presentative of the population
- Event POS: test is positive, Event DS: Down Syndrome present

$$\circ \quad P(\text{DS}|\text{POS}) = \frac{P(\text{DS} \cap \text{POS})}{P(\text{POS})} = \frac{\frac{48}{5282}}{\frac{1355}{5282}} = \frac{48}{1355}$$

Using conditional probability!

DS Status	Test Result		
	NEG	POS	Total
No DS	3921	1307	5228
DS	6	48	54
Total	3927	1355	5282

Independence and Conditional Probability

- Events A and B are independent if $P(A \cap B) = P(A)P(B)$
- By definition of conditional probability, $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)}$
- Thus, events A and B are independent if $P(A|B) = P(A)$
- Knowledge of B does not affect our opinion of how probable A is

Example 1: Cocaine Users

- Data collected in the US on cocaine usage
- Events A, B, C are disjoint
- $S = A \cup B \cup C$

Lifetime Use	Male	Female
1 – 19 times (A)	0.29	0.06
20 – 99 times (B)	0.16	0.18
100+ times (C)	0.23	0.08

Example 1: Cocaine Users

- Data collected in the US on cocaine usage
- Event M: user is male
- $P(M) = P(M \cap A) + P(M \cap B) + P(M \cap C) = 0.29 + 0.16 + 0.23 = 0.68$

Lifetime Use	Male	Female
1 – 19 times (A)	0.29	0.06
20 – 99 times (B)	0.16	0.18
100+ times (C)	0.23	0.08

06

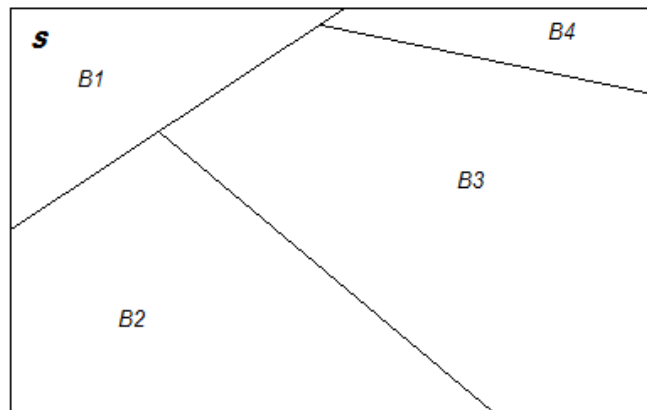
Law of Total Probability and Bayes' Theorem

Partition of the Sample Space



DEFINITION

$B_1, B_2, \dots, B_n$ are mutually exclusive events in the sample space S , and $B_1 \cup B_2 \cup \dots \cup B_n = S$.
Then $B_1, B_2, \dots, B_n$ are said to partition the sample space S .

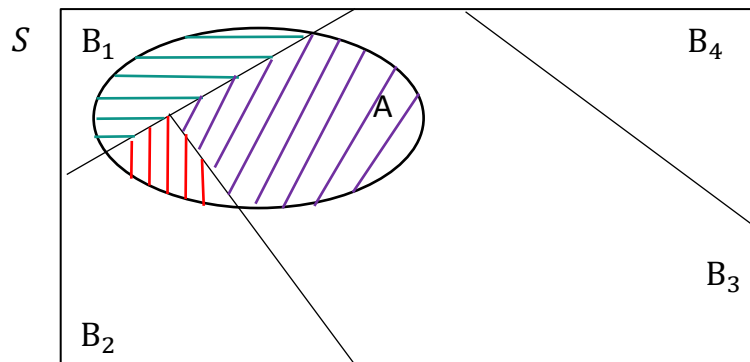


Law of Total Probability



DEFINITION

$$P(A) = \sum_{i=1}^n P(A \cap B_i)$$



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3)$$

Bayes Theorem

FORMULA

Events A and B are in the sample space S , where $P(A) > 0$.

$$P(B|A) = \frac{P(B)P(A|B)}{P(A)}$$

- $P(B|A) = \frac{P(B \cap A)}{P(A)}$
- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
 - $P(A \cap B) = P(B)P(A|B)$

Bayes Theorem

FORMULA

Events $B_1, B_2, \dots, B_n$ partition the sample space S . For any event A where $P(A) > 0$,

$$P(B_i|A) = \frac{P(B_i)P(A|B_i)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

$$P(B_i|A) = \frac{P(B_i \cap A)}{P(A)} = \frac{P(A|B_i)P(B_i)}{P(A)} = \frac{P(A|B_i)P(B_i)}{\sum_{i=1}^n P(A \cap B_i)} = \frac{P(A|B_i)P(B_i)}{\sum_{i=1}^n P(A|B_i)P(B_i)}$$

Example 4: Defective Products

- In a certain assembly plant, machines B_1 , B_2 and B_3 make 30%, 45% and 25% of the products respectively
- None of the products are made by any two machines
- 2%, 3% and 5% of products from B_1 , B_2 and B_3 are defective respectively
- If a product is randomly sampled, what is the probability it is defective?

Example 4: Defective Products

- S = sample space of all products
- Event A : random product is defective
- Event B_1, B_2 and B_3 : product is from machine B_1, B_2 and B_3 respectively
 - B_1, B_2 and B_3 partition S
 - B_1, B_2 and B_3 are disjoint and $B_1 \cup B_2 \cup B_3 = S$
- By Law of Total Probability,
$$\begin{aligned} P(A) &= P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3) \\ &= P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) \\ &= 0.02 \times 0.3 + 0.03 \times 0.45 + 0.02 \times 0.25 \\ &= 0.0245 \end{aligned}$$
- $$P(B_2|A) = \frac{P(A|B_2)P(B_2)}{P(A)} = \frac{0.03 \times 0.45}{0.0245} = 0.551$$

Sensitivity



DEFINITION

The sensitivity of a test is the probability that the test is positive, given that the person has the disease.

$$\text{Sensitivity} = P(+|D)$$

NOTATION

Event D : person has the disease

Event + : test is positive



KEY INSIGHT

A good quality test kit should have very high sensitivity.

False Positive



DEFINITION

False positive happens when the test shows positive for a non-disease subject.

$$P(\text{False positive}) = P(+|D^c)$$

NOTATION

Event D^c : person does not have the disease

Specificity



DEFINITION

The specificity of a test is the probability that the test is negative, given that the person does not have the disease.

$$\text{Specificity} = P(-|D^c)$$

NOTATION

Event D^c : person does not have the disease

Event $-$: test is negative



KEY INSIGHT

A good quality test kit should have very high specificity.

False Negative



DEFINITION

False negative happens when the test shows negative for a diseased subject.

$$P(\text{False negative}) = P(-|D)$$

NOTATION

Event D^c : person does not have the disease

Event $-$: test is negative

Prevalence



DEFINITION

Prevalence is the number of people who currently have the disease, divided by the number of people in the population.

$$\text{Prevalence} = P(D)$$

NOTATION

Event D : person has the disease

EXAMPLE

- The prevalence of diabetes is 11%
- If an individual is picked randomly, the probability this individual has diabetes is 11%

Example 5: Biomarker Assay

PROBLEM

A new biomarker diagnosis assay has sensitivity of 0.95 for a particular disease. It also has a specificity of 0.99 for the absence of the disease. The disease has prevalence of 0.005.

What is the probability that a person with a positive test actually has the disease?

GIVEN

- $P(+|D) = 0.95$
- $P(-|D) = 0.99$
- $P(D) = 0.005$

FIND

$$\begin{aligned} \bullet \quad P(D|+) &= \frac{P(+|D)P(D)}{P(+)} \\ &= \frac{P(+|D)P(D)}{P(+|D)+P(+|D^c)} \\ &= \frac{P(+|D)P(D)}{P(+|D)P(D)+P(+|D^c)P(D^c)} \\ &= \frac{0.95 \times 0.005}{0.95 \times 0.005 + (1-0.99) \times 0.995} \approx 0.323 \end{aligned}$$

Key Takeaways

1 Sample Space, Events and Axioms of Probability

2 Independence, Mutually Exclusive, Conditional Probability

3 Law of Total Probability and Bayes' Theorem

4 Epidemiological Terms